Performance Testing

Date	05 October 2026
Team ID	UNM-293 2026-2430
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Maximum Marks	5 Marks
Assigned To	Kaviya sree.T

Performance Testing

Performance testing evaluates how the FitBuddy system behaves under expected and peak load conditions. This document defines the performance-testing approach, test scenarios, target metrics, observations, and evidence requirements for the FastAPI + Gemini + SQLite based application.

Work Assigned to Kaviya sree.T: Execute the performance tests, record response-time/error/resource measurements, identify bottlenecks, and attach testing evidence/screenshots.

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter / Postman (or equivalent performance-testing tool)
Type of Testing	Load Testing, Stress Testing and basic Spike Testing
Target Module / API	/generate-workout, /submit-feedback, /view-all-users
Test Environment	Local FastAPI + Uvicorn server with SQLite database and Gemini API integration
Test Date	05 October 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Generate a personalized 7-day workout plan by submitting name, age, weight, goal and workout intensity.	5	60	Requests complete successfully with acceptable response time.
2	Submit feedback to revise an existing workout plan, such as adding cardio or rest days.	5	60	Updated plan is generated and returned without request failures.
3	Request nutrition/recovery tips based on the selected fitness goal using the Gemini Flash integration.	10	60	Tip requests remain responsive under concurrent load.
4	Access the admin view to retrieve registered users and original/updated plans.	10	60	User and plan data loads correctly without errors.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	To be measured	Pending	Record average after load test.
2	Response Time (Max)	< 5 seconds	To be measured	Pending	Record maximum response time.
3	Throughput (Req/sec)	To be measured	To be measured	Pending	Capture requests handled per second.
4	Error Rate	< 1%	To be measured	Pending	Record failed requests / total requests.
5	CPU Utilization	< 80%	To be measured	Pending	Monitor server CPU during test.
6	Memory Utilization	< 80%	To be measured	Pending	Monitor memory during test.

Step 4: Observations & Analysis

Key Findings:
Record the observed average/max response time, throughput, error rate, CPU utilization and memory utilization after executing the tests. The supplied project specification does not provide measured performance values, so actual results should be entered after testing.

Bottlenecks Identified:
Check for delays caused by Gemini API calls, concurrent request handling in FastAPI/Uvicorn, SQLite read/write operations, or resource saturation. Record only bottlenecks actually observed during testing.

Optimization Steps Taken:
Document any verified changes made to improve performance, such as reducing unnecessary database operations, improving request handling, or optimizing AI/API interaction.

Step 5: Screenshots / Evidence

Screenshot 1: Performance-testing tool summary / load-test results

Screenshot 2: API response / server resource monitoring evidence

Project Basis: The test scope is derived from the FitBuddy project specification, which describes personalized 7-day workout generation, feedback-based plan updates, nutrition/recovery tips, admin user/plan viewing, FastAPI routing, Gemini Pro/Flash integration, and SQLite/SQLAlchemy persistence.